SYMBIOSIS INSTITUTE OF TECHNOLOGY, PUNE

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Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

Assignment No. 13

Subject:	DevOps Lab
TCODE:	TE7950
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Branch	CSE, Batch (2023-27), L3
Academic Year & Semester	2025 – 2026, Semester VII
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Title of Assignment:	Kubernetes Service & Networking

Theory:	A Kubernetes Service is an abstraction that provides a stable network endpoint for accessing one or more Pods in a Kubernetes cluster. Since Pods are temporary and their IP addresses may change, a Service uses labels and selectors to route incoming traffic to the appropriate Pods. A NodePort Service exposes the application outside the cluster by allocating a port on each cluster node, allowing external access to the application. During local development, kubectl port-forward can also be used to securely forward traffic from a local machine to a Kubernetes Service for testing and verification. This approach simplifies communication between applications and ensures reliable access to containerized workloads. **Commands Used** • Created the Kubernetes Deployment: kubectl apply -f web-deployment.yaml • Verified that the Pods were running: kubectl get pods • Created the Kubernetes Service: kubectl apply -f web-service.yaml • Displayed the available Kubernetes Services: kubectl get service • Displayed detailed information about the Service: kubectl describe service web-app-service • Displayed the Service endpoints: kubectl get endpoints

	• Displayed the running Pods with labels: kubectl get pods --show-labels • Displayed the application logs: kubectl logs web-app-deployment-<pod-name> • Forwarded the Kubernetes Service to the local machine: kubectl port-forward service/web-app-service 8085:80 • Deleted the Kubernetes Service: kubectl delete -f web-service.yaml • Deleted the Kubernetes Deployment: kubectl delete -f web-deployment.yaml

Implementation:

Step 1: Create the Kubernetes Service YAML

DevOps-Assignments > Assignment-13-Kubernetes-Service > k8s-manifests > ! web-service.yaml > ...

```yaml
apiVersion: v1
kind: Service
metadata:
  name: web-app-service
spec:
  type: NodePort
  selector:
    app: web-app
  ports:
    - port: 80
      targetPort: 5001
      protocol: TCP
```

Created Kubernetes Service Manifest — Created the web-service.yaml file defining a Kubernetes NodePort Service to expose the Flask web application running in the Kubernetes Pods.

Step 2: Apply the Service

```
(base) pushkrajpnaik@Pushkrajs-Laptop Assignment-12-Kubernetes-Deployment % kubectl get deployments
kubectl get pods
NAME                READY   UP-TO-DATE   AVAILABLE   AGE
web-app-deployment   2/2     2            2           85s
NAME                                  READY   STATUS    RESTARTS   AGE
web-app-deployment-684bc49f59-t7bkl   1/1     Running   0          85s
web-app-deployment-684bc49f59-wcp56   1/1     Running   0          85s
(base) pushkrajpnaik@Pushkrajs-Laptop Assignment-12-Kubernetes-Deployment % kubectl port-forward deployment/web-app-deployment 8085:5001
Forwarding from 127.0.0.1:8085 -> 5001
Forwarding from [::1]:8085 -> 5001
Handling connection for 8085
Handling connection for 8085
^C
```

Applied Kubernetes Service — Applied the Kubernetes Service manifest using kubectl apply, creating a NodePort Service for the deployed Flask application.

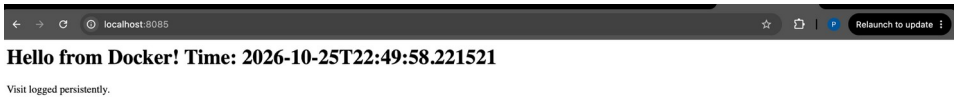
Step 3: Find the Service Information

```
(base) pushkrajpnaik@Pushkrajs-Laptop Assignment-13-Kubernetes-Service % kubectl get service
NAME              TYPE        CLUSTER-IP     EXTERNAL-IP   PORT(S)        AGE
kubernetes        ClusterIP   10.96.0.1      <none>        443/TCP        35m
web-app-service   NodePort    10.96.13.75    <none>        80:32349/TCP   2m51s
```

Verified Kubernetes Service — Verified the Kubernetes NodePort Service and identified the assigned NodePort (30649) used to expose the Flask web application outside the Kubernetes cluster.

Step 4: Access the Application





Accessed Application Through Kubernetes Service — Used kubectl port-forward to forward the Kubernetes Service to the local machine and successfully accessed the Flask web application at http://localhost:8085, verifying that it was running correctly.

Step 5: Clean Up



Deleted Kubernetes Service — Removed the Kubernetes Service using the kubectl delete command, cleaning up the networking resources created for the application.

Subject Faculty Name: Prof. Arijit Dutta